

FORD FULKERSON ALGORITHM

Problem

Submissions

Leaderboard

Discussions

Submitted 4 days ago • Score: 5.00

Status: Accepted



Test Case #0

Submitted Code

Language: Java 15

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```
1 import java.io.*;
2 import java.util.*;
3
4 public class Solution {
5
6     static int V;
7
8     static boolean bfs(int[][] residual, int source, int sink, int[] parent) {
9         boolean[] visited = new boolean[V];
10        Queue<Integer> queue = new ArrayDeque<>();
11
12        queue.add(source);
13        visited[source] = true;
14        parent[source] = -1;
15
16        while (!queue.isEmpty()) {
17            int u = queue.poll();
18
19            for (int v = 0; v < V; v++) {
20                if (!visited[v] && residual[u][v] > 0) {
21                    parent[v] = u;
22                    visited[v] = true;
23
24                    if (v == sink)
25                        return true;
26
27                    queue.add(v);
28                }
29            }
30        }
31
32        return false;
33    }
34
35    static int maxFlow(int[][] graph, int source, int sink) {
36        int[][] residual = new int[V][V];
37
38        for (int i = 0; i < V; i++) {
39            residual[i] = graph[i].clone();
40        }
41
42        int[] parent = new int[V];
43        int maxFlow = 0;
```

```
44
45     while (bfs(residual, source, sink, parent)) {
46         int pathFlow = Integer.MAX_VALUE;
47
48         for (int v = sink; v != source; v = parent[v]) {
49             int u = parent[v];
50             pathFlow = Math.min(pathFlow, residual[u][v]);
51         }
52
53         for (int v = sink; v != source; v = parent[v]) {
54             int u = parent[v];
55
56             residual[u][v] -= pathFlow;
57             residual[v][u] += pathFlow;
58         }
59
60         maxFlow += pathFlow;
61     }
62
63     return maxFlow;
64 }
65
66 public static void main(String[] args) throws Exception {
67     BufferedReader br =
68         new BufferedReader(new InputStreamReader(System.in));
69
70     StringTokenizer st =
71         new StringTokenizer(br.readLine());
72
73     V = Integer.parseInt(st.nextToken());
74     int E = Integer.parseInt(st.nextToken());
75
76     int[][] graph = new int[V][V];
77
78     for (int i = 0; i < E; i++) {
79         st = new StringTokenizer(br.readLine());
80
81         int u = Integer.parseInt(st.nextToken());
82         int v = Integer.parseInt(st.nextToken());
83         int capacity = Integer.parseInt(st.nextToken());
84
85         graph[u][v] += capacity;
86     }
87
88     int source = 0;
89     int sink = V - 1;
90
91     System.out.println(maxFlow(graph, source, sink));
92 }
93 }
94
```